Grid optimization using DER grid support functions

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***Abstract*— As technology makes significant advancements in every field, the conventional power grid management struggles to keep up with the increasing needs of the modern society. Growing cities and constantly increasing number of power consumers, mandate the round the clock need for power. But there are more variables in this equation. The fuel reserves needed for conventional power plants are slowly, but inevitably depleting. The CO2 emissions grow and need to be reduced as well due to the impact on the global climate. Therefore, countries all over the world introduced highly ambitious targets for reduction of carbon emissions by incentivizing the adoption of new technologies for producing electric power.**

All these are the reasons for a significant increase of the renewable energy sources, even on the consumer side in the power distribution grids. In order to optimally use the potential of those resources for the benefit for both customers and the distribution network to which they are connected, these resources need to be managed intelligently and in coordination with distribution network conditions. Inadequate use of distributed energy resources (DERs) can easily cause problems in operating the power grid in a reliable and secure way and the customer could be limited and restricted due to inability of the power grid to deliver energy produced by the customers. This is another reason for the need of intelligent management of DERs. Therefore, a number of standards have been imposed, mandating what requirements DERs must meet. These standards encompass the grid support functions a DER should implement in order to avoid adverse DER impact on power grid operations.

If there is a very large number of DERs, including customer-site DERs, it is technically and economically hard to monitor and control them all using standard SCADA protocols. Therefore, different protocols for massive and cost-effective implementation were introduced. One of those is an IEEE 2030.5 protocol, built for internet-based communication, used to manage DERs across the utility service area. The IEEE 2030.5-2018 has recently been updated to incorporate the IEEE 1547-2018 functionality related to grid support functions in the standard. In order to operate distribution network in an efficient and reliable way, it is crucial to coordinate DER grid support functions with a distribution system conditions and objectives trough centralized system as ADMS and/or Utility DERMS.

Keywords—IEEE 2030.5, Distribution network, Prosumer, Distribution Generation, Distributed Energy Resources, Advanced Distribution Management System, Smart inverter functions.

I. INTRODUCTION

Renewable power generators, energy storage, electric vehicles, controllable demand, known collectively as DER, have made it feasible for customers to use these technologies to locally generate, store, and manage power at their premises. These DER are often connected to the power grid at the distribution level, where their presence in large scale can be disruptive for the distribution network and challenges the operational actions to plan, design, operate and manage distribution network to ensure reliable and quality supply to each customer. They introduce two-way power flow at locations through the distribution network, where renewable energy, intermittent by nature, additionally creates challenge to ensure reliable and safe services to all customers [1]-[4]. On the other side, DER provide new opportunities for utilities to utilize all advantages that DER offer for managing the grid. There is an ongoing effort to adjust the technical operating standards for DER interconnection across the globe in order to accommodate the power flows from DER, thus minimizing adverse impact on distribution grid operation. Interconnection standards, such as IEEE 1547 [5], propose set of functions and capabilities that DER must support in order to help distribution grid operations to better cope with new distribution grid conditions with higher penetration of DER on certain grid circuits, including those on LV side. Instead of restricting the new DER connection to prevent possible adverse impact of DER, interconnection standards recommend and propose functions that DER will perform to support power system operations. In most cases, DER are connected to the grid by smart inverters, the devices that are highly capable of supporting grid operation through configuration of set of functionalities such that they can respond quickly to the local conditions and/or remote commands. Standardization bodies regularly update interconnection requirements as well as operational guidelines. In particular, the Smart Inverter Working Group (SIWG) developed the grid support functions which allow DER's to modify their

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parameters according to locally observed conditions. These functions were integrated into Common Smart Inverter Standard (CSIP) [6] and utilize IEEE2030.5 as its communication protocol.

Smart inverters can operate autonomously by adjusting DER's real and reactive power output as a response to a local condition. Autonomous operation is enabled by specifying smart inverter setting which defines the operating conditions. For instance, smart inverter can monitor local voltage and respond accordingly to the measured voltage value by adjusting the reactive power output and thus helping maintain local voltages in allowed ranges. This type of operation can be defined by smart inverter's Volt VAR curve setting.

This paper describes autonomous inverter functionalities as well as advanced inverter functionalities and the way they are considered in centralized ADMS (or Utility DERMS) solution to improve the management of distribution network condition in high DER penetration.

II. Centralized and local control

If there is a very large number of DER, including customer-site DER, it is technically and economically hard to monitor and control them all. Inadequate use of DER can easily cause problems in operating the distribution network in a reliable and secure way and the customer could be limited and restricted due to inability of the distribution network to deliver energy produced by the customers. These are reasons for the need of intelligent management of DER.

Smart inverters support a number of advanced functionalities which can help utility to operate and optimize distribution network by utilizing smart inverter as controllable resources as well.

If smart inverters are directly monitored and controlled by ADMS in real time, they can respond to commands issued by ADMS. In general, there are two ways how they can be controlled by ADMS:

- Issuing an emergency command which overrides the autonomous smart inverter operations,
- Updating the smart inverter setting (e.g. adjust schedule of energy storages for energy or ancillary services or adjusting Volt-VAR curve in accordance to desired network optimization objective) in order to coordinate autonomous smart inverter operations with centralized ADMS applications' objectives.

If DER capabilities can be properly modelled and integrated into a centralized ADMS system, high DER penetration can be transformed from problematic uncertainties to a beneficial tool for distribution network management.

By enabling smart inverter functionalities, DER support network operation and improve the network performances locally. However, if they are managed on an aggregated level and coordinated with other traditional network assets, they can improve overall network performances on both circuit and system level and help utility to optimally operate the distribution network and achieve overall benefits through reduced losses and improved reliability.

Therefore, modeling and integration of smart inverter functionalities into ADMS provides a solution which combines centralized and decentralized (local autonomous) control of traditional network assets and DER in a coordinated way.

Modeling and simulating autonomous inverter operations is beneficial and important for updating the status and output of DER in ADMS in accordance to inverter operations in the field in real time as well as to evaluate effect of suggested operations on inverter operations and consequently on the network state (network loading and voltages). In order to avoid inconsistence between unmonitored DER status in the field and its status in ADMS, ADMS should estimate status and output of DER based on the inverter operation logic.

Advanced inverter functionalities enable remote DER usage. Utilizing DER along with traditional distribution resources to operate and optimize the network improves system reliability. For instance, DER utilization minimizes operation of other distribution resources and thus prolongs their life time (i.e. minimizing the switching of capacitors for VAR support). Moreover, utilizing DER increases the ability to find solution for network operation and optimization.

Through smart inverter functionalities, DER become new network resources with Watt and VAR control capability. Leveraging the control of DER along with traditional network resources (tap changers, capacitors, regulators, switching devices) in a synchronized fashion would help the utility to maintain optimal voltages, and ensure network operation within operational limits.

III. Communication with DER using IEEE 2030.5 protocol

ADMS system needs to be aware of DER, their location in the distribution network, and how they can be configured so that their effect on the distribution network is optimal at all times. Interconnection standards (grid codes) define the functionalities a DER must support to be connected to the distribution network, while communication protocols standardize the monitoring and control interface for these DER. Regional regulations, membership, or cooperation with different standardization groups, as well as the choice of the equipment vendor, all affect the standards a utility adopts and expects a utility ADMS solution to support.

An ADMS model first needs to add DER with all their technical and contractual parameters, usually coming from the GIS. Then, the ADMS needs to interface with these devices, which are connected to the distribution network.

SCADA communication protocols such as DNP3.0 and IEC 61850 are adopted by utilities worldwide and have been used to interface with the traditional devices such as circuit breakers, load break devices, LTCs, voltage regulators, capacitors and different types of sensors that are installed in the distribution network and owned by the utility, for the purpose of providing safe, reliable and quality electric power to the end customers.

However, monitoring and control of DER is different in nature and SCADA communication is not applicable in a majority of cases for the following reasons:

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Time dimension - Traditional devices are available in real-time and can provide their function (e.g. change network configuration, provide voltage regulation) almost always when the ADMS issues a command. The availability of DER is much more complex than the availability of traditional devices. It is linked to the interests of diverse DER owners, as well as DER specific technical constraints. The result is that time dimension must be considered as well to account for and ensure DER availability:

- DER is available only at certain time (e.g. an e-vehicle can be charged at night),
- To be available, DER requires an upfront request (DER programs it can participate in),
- DER can offer services to various parties. Once it offers service to one party, it is unavailable for usage by another party, e.g. DER that participate in the markets that are agreed on the day-ahead cannot be used for service in the distribution network.
- Energy storages must charge and discharge regularly to preserve their capacity and effectiveness. To be used at the time of need, energy storages must have an adequate state of charge at that moment that must be prepared upfront.

To dispatch DER that need an upfront dispatch request, the system must be aware of the need for their dispatch ahead of time. Thus, it is necessary for utility system to also dispatch scheduled output of DER and for communication protocol to communicate this request to DER device on appropriate way.

Expensive SCADA implementation - Majority of DER are residential, behind-the-meter, small-size units connected to low voltage distribution network and owned by end customers, like roof-top solar, air-conditioning appliances, electric vehicles, home batteries, etc. It is not economical to connect every residential DER with a SCADA protocol that requires a dedicated RTU that can be expensive for residential DER owners. However, the ability of residential DER to connect to the internet creates a way and need for communication protocol that runs on the internet. Internet connectivity, however, introduces cyber-security challenges that protocol standardization must account for.

Aggregation – A small DER creates local benefit to its owner. Only aggregated small DER can benefit to utility in managing network constraints and providing services to manage demand on different level of network hierarchy. Complexity of monitoring and dispatch of the aggregated units is mostly related to the fact that the exchanged data is on the aggregated level, while the local impact of such data must be supposed at some, good enough, level. The communication interface must include information about the aggregation point (e.g. transformer, feeder), but also the types of service these aggregated units can provide so that the utility DERMS can utilize them.

These introduced new aspects that need to be considered and addressed by the communication protocols specified for DER.

The most advanced protocol, built for internet-based communication, used to manage DER across the utility service territory is IEEE 2030.5. IEEE 2030.5-2018 has recently been updated to incorporate the IEEE 1547-2018 functionality in the standard. It is based on web services with built-in security and is designed to use the internet for interface between the system and devices.

IEEE 2030.5 standard is in its initial stages of implementation worldwide. As a robust, all-encompassing standard covering a multitude of industries and use cases, it is often implemented as a subset (profile) of interest for the particular region or industry. Its most advanced implementation as of 2020 is California Smart Inverter Profile (CSIP) v2.0. It is mandated as the preferred industry standard for DER communications in California and is likely to be used as the guideline for other worldwide profiles and implementations as well.

This protocol is based on the most advanced DER data model representing all the technical and contractual parameters of DER, along with control, scheduling and reporting data:

- DER capability,
- DER settings,
- DER status,
- DER availability,
- DER monitoring,
- DER control and scheduling, and
- DER reporting.

This data model has been built to cover the range from the smallest DER to an aggregator head-end. It is well suited to prosumers and aggregators, often with several layers of aggregation and thus expands the range of devices that can be monitored and controlled.

The large-scale DER such as solar and energy storage or combined solar and storage system can be considered as real-time controllable resource to control power exchanged between the network and DER across the point of common coupling. The large-scale DER will still be able to use standard SCADA real-time protocols, when affordable. IEEE 2030.5 could be an alternative to SCADA protocols in some circumstances.

There are two scenarios of communication between utility head end server and DER systems. The two scenarios envisioned for communications between the utility server and DER systems are Direct DER Communications and Aggregator Mediated Communications. In both cases, the communications path to the utility is governed by regulatory and utility requirements, and the IEEE 2030.5 protocol.

In the Direct DER Communications scenario, the utility communicates with the DER system directly.

In the Aggregator Mediated Communications scenario, the utility communicates with an aggregator back end management

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system rather than directly with individual DER. The aggregator is assumed to be managing a fleet of inverters that are distributed across the utility's service territory rather than having a single point of common coupling. The aggregator is then responsible for relaying any requirements for DER operational changes or data requests to the affected systems and returning any required information to the utility. Each DER controlled by the aggregator appears as an IEEE 2030.5 EndDevice to the utility server. The likely applicability is for fleet operators and aggregation service providers.

By supporting functionalities specified in interconnection standards, DER become capable to autonomously operate by adjusting its status, active and reactive power as a response to local conditions and/or to be controlled and managed by the utility (network operator). The autonomous functions which are recommended by the IEEE 2030.5 to support network operation and improve the network performances locally are:

- Low/High Voltage Ride Through
- Low/High Frequency Ride Through
- Ramp Rate Setting
- Connect/Disconnect
- Dynamic Volt-VAr
- Fixed Power Factor Control
- Real Power Output Limit Control
- Volt-Watt Control
- Frequency-Watt Control
- Set Active Power Mode (in percentage of Max power or in Watts) [6]

IV. Using Dynamic Volt/VAR Function for Coordinated Central and Local Voltage Management

Dynamic Volt/VAR function allows DER to respond to voltage deviation from the target voltage value by consuming or producing certain amount of reactive power. In other words, dynamic Volt/VAR curve defines the changes in DER's VAR output in response to the changes in the measured voltage at the point of DER connection to the distribution network [7].

The target amount of reactive power output is specified by a curve defining dependency of reactive power output on voltage value, as presented in Fig 1. This function should ensure that DER consumes reactive power (under excitation) in response to an increase in voltage and produces reactive power (overexcitation) in response to a decrease in voltage. While the voltage is within deadband around referent (target) voltage value, reactive power output is zero [7].

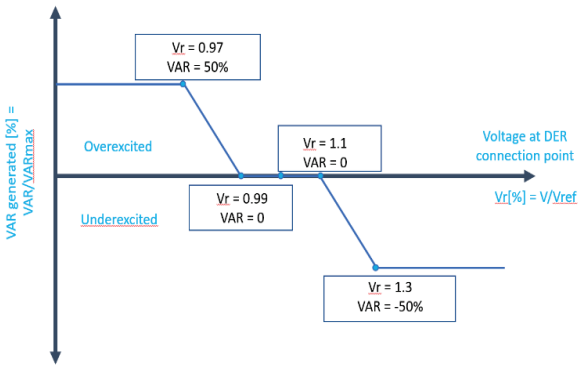


Fig 1 – Dynamic Volt VAR curve

The purpose of Volt VAR function is to help maintain voltages within the allowed ranges. Configuring this capability at the point of DER connection will ensure DER maintain voltages at the point of connection and consequently along the feeder where it is connected. This capability can be important for maintaining voltages above low limit on a long feeder, where setting the higher target value at DER location will ensure voltages above low limit at the end of long circuits. Generally, higher setting of target voltage value will ensure acceptable voltage profile along the feeder, however, in the same time, it could minimize or restrict energy efficiency and power reduction benefit that could be provided from voltage management. Power reduction and energy efficiency could be achieved through voltage management by operating tap changers to reduce voltages along the feeders while keeping them above acceptable low limit at customer connection points.

Let's consider one long feeder, illustrated in Fig 2, that is monitored and operated by centralized ADMS solution. There is a DER connected at the middle of the feeder with configured Volt VAR curve to maintain voltage at point of connection at the value around 225 V (normalized value for LV voltage / customer voltage value) to ensure good customer voltages along the feeder.

Centralized ADMS solution is configured to optimize network and DER in order to achieve power reduction. The diagram shown in Figure 3 illustrates voltage profile along the feeder and tap changer and DER impact on the voltage profile (green dashed line). As it is presented in the diagram, by only lowering tap changers at the transformer that supplies the feeder from Position 1 to Position 2, the system will not achieve full benefits, as DER operates per predefined Volt VAR curve and raises the voltage at the point of connection which will keep the voltage downstream the feeder at almost the same value (green dashed line) regardless of the tap changer position.

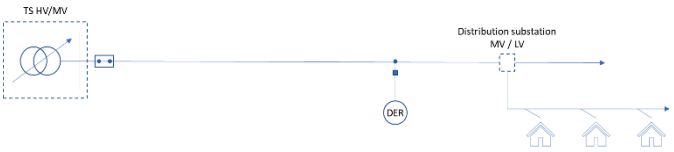


Fig 2 - Long MV feeder

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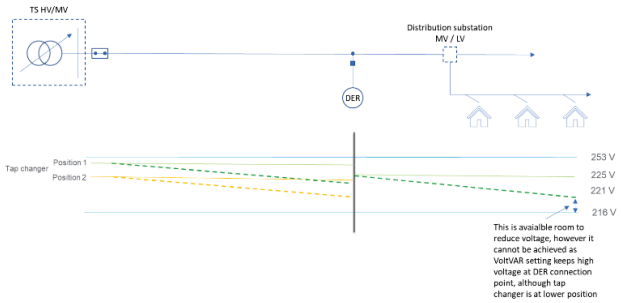


Fig 3 - Feeder voltage profile for different tap charger position – Limited power reduction benefit

Centralized ADMS solution is configured to optimize network and DER in order to achieve power reduction. The diagram shown in Fig *3* illustrates voltage profile along the feeder and tap changer and DER impact on the voltage profile (green dashed line). As it is presented in the diagram, by only lowering tap changers at the transformer that supplies the feeder from Position 1 to Position 2, the system will not achieve full benefits, as DER operates per predefined Volt VAR curve and raises the voltage at the point of connection which will keep the voltage downstream the feeder at almost the same value (green dashed line) regardless of the tap changer position.

To achieve full benefit, the centralized system should adjust not just the tap changer position, but also Volt VAR configuration setting of DER as it is presented in Fig *4*. The target voltage of initial Volt VAR configuration was set up at 225 V (normalized value at LV voltage / customer voltage value). However, the system proposed target voltage change to 221 V which will result in lower voltage values downstream the feeder, while keeping them above low acceptable limit.

When dynamic Volt VAR function is enabled on DER's smart inverter, reactive power output of DER depends on predefined dynamic Volt VAR settings and local voltage conditions at the point of DER connection to the distribution network.

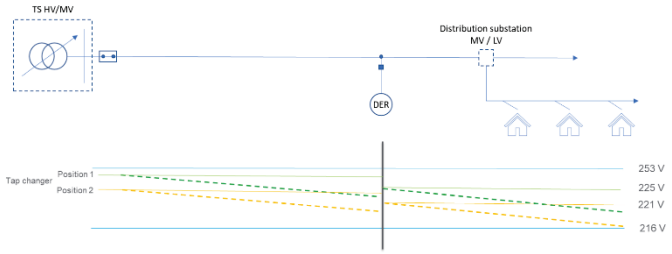


Fig 4 - Feeder voltage profile for different tap charger position and adjusted Volt VAR setting – Full power reduction benefit

Modeling operation of dynamic Volt VAR curve in centralized solution that is responsible for managing network operation provides:

- Estimation of reactive power output considering Volt VAR curve and calculated line voltage at the point of DER connection to the distribution network. The effect of Volt VAR curve will be taken into calculations to determine reactive power output and its impact on network state.
- Adjustment of Volt VAR curve setting through centralized system to achieve some of required objectives such as power consumption reduction or conservation voltage reduction or requested VAR support on large scale level.

V. Conclusion

Centralized ADMS systems are still used to manage the distribution network. DER can contribute greatly to the status of the network. However, the full benefit for network operation cannot be accomplished with just autonomous operations of DER. To achieve that, advanced algorithms based on the knowledge of the entire distribution network must be utilized. ADMS is a system that has the knowledge of those algorithms and can utilize them, but in order to do so, sufficient data must be provided to and by ADMS. The DER involved must be able to provide and receive this data as well. Very large number of these resources, as well as the manufacturer diversity, mandate the standardization of the communication protocols. The data needed for optimal utilization of distribution network resources can then be exchanged efficiently and securely between ADMS and DER. Investments must be made in the standardization of the DER clients as well as head end servers and furthermore in their integration with ADMS systems.

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